

Unbiased exact diagonalization of quantum spin ice in the π -flux regime: removing finite-size contamination by translationally-invariant twist averaging

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Abstract

The thermodynamics and dynamics of quantum spin ice (QSI) are governed by an emergent compact $U(1)$ gauge field. Its unfrustrated 0-*flux* sector ($J_{\pm} > 0$) is accessible to sign-problem-free quantum Monte Carlo (QMC); the frustrated π -*flux* sector ($J_{\pm} < 0$) realized in the dipolar-octupolar cerium pyrochlores carries a QMC sign problem, leaving exact diagonalization (ED) as the only unbiased finite-temperature method. ED has a finite-size problem specific to QSI: a periodic cluster hosts short ice-preserving ring exchanges that wind the torus and have no infinite-lattice analogue. The dominant ones are four-site loops with coupling $g_4 = 4J_{\pm}^2/J_{zz}$, which is *second* order in $J_{\pm}$ and hence parametrically larger than the genuine photon coupling $g_{\text{hex}} = 12|J_{\pm}|^3/J_{zz}^2$; they contaminate exactly the emergent-gauge observables (the low- T specific-heat peak and the low-energy $S^{zz}(\mathbf{q}, \omega)$). We remove this contamination by *twist averaging in the translationally-invariant gauge*: threading a $U(1)$ flux through the cluster as a uniform vector potential on *every* bond and averaging the effective Hamiltonian over the flux. We show, mathematically and by direct computation, that this average annihilates every winding ring exchange while preserving every contractible one – provided two conditions are met that the naive procedure violates: the flux must be distributed uniformly (the boundary-only gauge removes merely half), and the average must span the doubled period set by the half-integer transport of the emergent charges (the naive 2π period leaves a 40% residual). Both conditions have a transparent origin: a winding ring exchange transports emergent charge across the cluster, and only a translationally-invariant flux, averaged over its true period, dephases that transport completely. On the 16- and 32-site clusters the method moves the low- T gauge peak and the low-energy S^{zz} weight from the artifact scale g_4 to the photon scale g_{hex} for both flux signs, while leaving the high- T charge (spinon) peak – genuine, uncontaminated physics – untouched. This makes a 32-site pyrochlore an unbiased laboratory for the π -flux gauge sector, to be benchmarked against 0-flux QMC and applied at the exchange parameters of the cerium pyrochlores.

1 The problem

Quantum spin ice and its emergent photon. The minimal model of QSI is the XXZ antiferromagnet on the pyrochlore lattice, in the local $\langle 111 \rangle$ frames,

$$H = J_{zz} \sum_{\langle ij \rangle} S_i^z S_j^z - J_{\pm} \sum_{\langle ij \rangle} (S_i^+ S_j^- + S_i^- S_j^+), \quad 0 < |J_{\pm}| \ll J_{zz}. \quad (1)$$

For $J_{\pm} \rightarrow 0$ the ground manifold is the exponentially degenerate set of 2-in-2-out (ice-rule) states. A small $J_{\pm}$ lifts the degeneracy through virtual ring exchanges and generates the deconfined $U(1)$ lattice gauge theory of QSI: S^z is the emergent electric field, the third-order hexagonal ring exchange

$$g_{\text{hex}} = \frac{12|J_{\pm}|^3}{J_{zz}^2} \quad (2)$$

is the Maxwell term whose quantum is the emergent photon, and ice-rule defects are gapped, fractionalized spinons (pair threshold $\Delta_s \approx 0.8 J_{zz}$ on the clusters below). The *sign* of the ring coupling follows $\text{sgn}(J_{\pm}^3)$: $J_{\pm} > 0$ gives the unfrustrated 0-flux vacuum, $J_{\pm} < 0$ the frustrated π -flux vacuum. Thermodynamically the photon appears as a low- T Schottky peak in $C(T)$ at $T \sim g_{\text{hex}}$, far below the spinon (charge) peak at $T \sim J_{zz}$; dynamically it appears as low-energy weight in the non-spin-flip channel $S^{zz}(\mathbf{q}, \omega)$.

Why π -flux needs exact diagonalization. In the S^z basis the off-diagonal matrix elements of eq. (1) are $-J_{\pm}$. For $J_{\pm} > 0$ they are negative and worldline/stochastic-series QMC is sign-problem-free: the 0-flux QSI is quantitatively characterized. For $J_{\pm} < 0$ they are positive and QMC acquires a sign problem. The π -flux regime – believed to describe $\text{Ce}_2\text{Zr}_2\text{O}_7$, $\text{Ce}_2\text{Sn}_2\text{O}_7$, $\text{Ce}_2\text{Hf}_2\text{O}_7$ – is thus a QMC dead zone, and unbiased finite- T information must come from exact diagonalization of a finite cluster.

The finite-size problem. A periodic cluster is a flat torus, and its periodic identifications generate ice-preserving ring exchanges that *wind* the torus with no counterpart on the infinite lattice. The shortest and most damaging are *four-site* loops. Because a four-site loop factorizes into two disjoint nearest-neighbour (NN) hops, it is generated at *second* order in $J_{\pm}$,

$$g_4 = \frac{4J_{\pm}^2}{J_{zz}}, \quad \text{so} \quad \frac{g_4}{g_{\text{hex}}} = \frac{J_{zz}}{3|J_{\pm}|} \gg 1. \quad (3)$$

The artifact is thus *parametrically larger* than the physics, and grows as $|J_{\pm}| \rightarrow 0$ – the weak-coupling regime of the cerium materials. It lives entirely in the ice manifold, so it contaminates precisely the emergent-gauge observables: it pushes the low- T $C(T)$ peak and the low-energy S^{zz} weight up to the scale g_4 instead of g_{hex} (quantified in section 6). The spinon continuum (S^{+-} at $\omega \sim J_{zz}$) lies outside the ice manifold and is *not* contaminated.

Goal. Remove this finite-size contamination from the gauge sector of a periodic-cluster ED, so that the 32-site pyrochlore delivers unbiased π -flux thermodynamics and dynamical structure factor.

2 The effective gauge Hamiltonian and the winding loops

Cluster torus. The cluster carries N sites with periodic identifications generated by the rows of an integer matrix $\mathbf{L} \in \mathbb{Z}^{3 \times 3}$ (in units of the cubic lattice constant a); position space is the flat torus $T^3 = \mathbb{R}^3 / \mathbf{L}\mathbb{Z}^3$ with first homology $\mathbb{Z}^3$. Each NN bond (i, j) has a *wrap vector* $\mathbf{n}_{ij} = \text{round}[(\mathbf{r}_j - \mathbf{r}_i)\mathbf{L}^{-1}]$, zero in the interior and nonzero across the boundary.

Effective Hamiltonian. Degenerate $(J_{\pm})$ -perturbation theory in the ice manifold yields $H_{\text{eff}} = \sum_{\mathcal{C}} c_{\mathcal{C}} O_{\mathcal{C}}$, a sum of ring-exchange operators $O_{\mathcal{C}} = (S_a^+ S_b^- S_c^+ S_d^- \dots)$ over ice-preserving cycles $\mathcal{C}$; because each intermediate resolvent is Ising ($\propto J_{zz}$), $c_{\mathcal{C}} \propto J_{\pm}^{|\mathcal{C}|/2}$ exactly. On the $2 \times 2 \times 2$ FCC cluster

Table 1: Census of ice-preserving ring exchanges. “Winding/wrapping” loops close only through the periodic boundary ($\mathbf{w} \neq \mathbf{0}$); “contractible” hexagons are genuine bulk plaquettes ($\mathbf{w} = \mathbf{0}$). The last column is the survival fraction under translationally-invariant twist averaging (section 3).

cluster	channel	count	order	kept by the average
$1 \times 1 \times 1$ cubic ($N=16$)	winding four-loops	36	$J_{\pm}^2$	0
	wrapping hexagons	48	$J_{\pm}^3$	0
	contractible hexagons	16	$J_{\pm}^3$	1
$2 \times 2 \times 2$ FCC ($N=32$)	winding four-loops	24	$J_{\pm}^2$	0
	wrapping hexagons	96	$J_{\pm}^3$	0
	contractible hexagons	32	$J_{\pm}^3$	1

($N = 32$) the ice manifold has dimension 2970, so H_{eff} is *fully diagonalizable* and its thermodynamics is numerically exact. Enumeration (Schrieffer–Wolff with per-process bookkeeping) gives table 1: the winding four-loops and wrapping hexagons are the contamination; the contractible hexagons carry the photon.

3 The method: twist averaging in the translationally-invariant gauge

Twist as a flat $U(1)$ connection. Thread a $U(1)$ flux $\varphi \in [0, 2\pi)^3$ through the three non-contractible cycles by attaching a phase to each transverse hop. The gauge-invariant content of the flux is its holonomy $e^{-i\mathbf{w} \cdot \varphi}$ around a loop of winding $\mathbf{w}$, but *how* the phase is distributed over bonds is a gauge choice, and the choice matters for the average below. We use the *translationally-invariant* (“smooth”) gauge – a uniform vector potential $\mathbf{A} = \mathbf{L}^{-1}\varphi$ that places $\theta_{ij} = \mathbf{A} \cdot \mathbf{d}_{ij}$ on *every* bond, with $\mathbf{d}_{ij}$ the minimum-image bond displacement:

$$H_{\text{eff}}(\varphi): \quad c_c \rightarrow c_c e^{-i\mathbf{A} \cdot \delta_c}, \quad \delta_c = \sum_{\text{hops}} \mathbf{d} = \underbrace{\sum_{i \in R} \mathbf{r}_i - \sum_{j \in L} \mathbf{r}_j}_{\boldsymbol{\rho}} + \mathbf{N} \cdot \mathbf{L}, \quad (4)$$

where δ_c is the *net charge transport* of the process – the physical displacement of the moved S^z (boson) density, $\boldsymbol{\rho}$ its intrinsic dipole and $\mathbf{N} \cdot \mathbf{L}$ the wrap contribution. Thus in the smooth gauge each ring exchange is weighted by a phase set by how far it transports emergent charge across the cluster (fig. 1).

The average. The method is to average the *Hamiltonian* over the flux,

$$\overline{H_{\text{eff}}} = \int H_{\text{eff}}(\varphi) \frac{d\varphi}{(\text{period})^3}, \quad c_c \rightarrow c_c \overline{e^{-i\varphi \cdot (\mathbf{L}^{-1}\delta_c)}}. \quad (5)$$

A process survives iff its transport phase averages to unity, i.e. iff $\mathbf{L}^{-1}\delta_c \in \mathbb{Z}^3$, i.e. iff $\delta_c \in \mathbf{L}\mathbb{Z}^3$. Two points, elaborated and proved in sections 4 and 5, are essential and are where the naive procedure fails:

- (i) *Translational invariance.* The phase must sit on every bond. In the boundary-only gauge a winding loop has a virtual path built from interior bonds alone, which carries no phase and survives the average – leaving half the coupling. The uniform gauge closes this loophole.

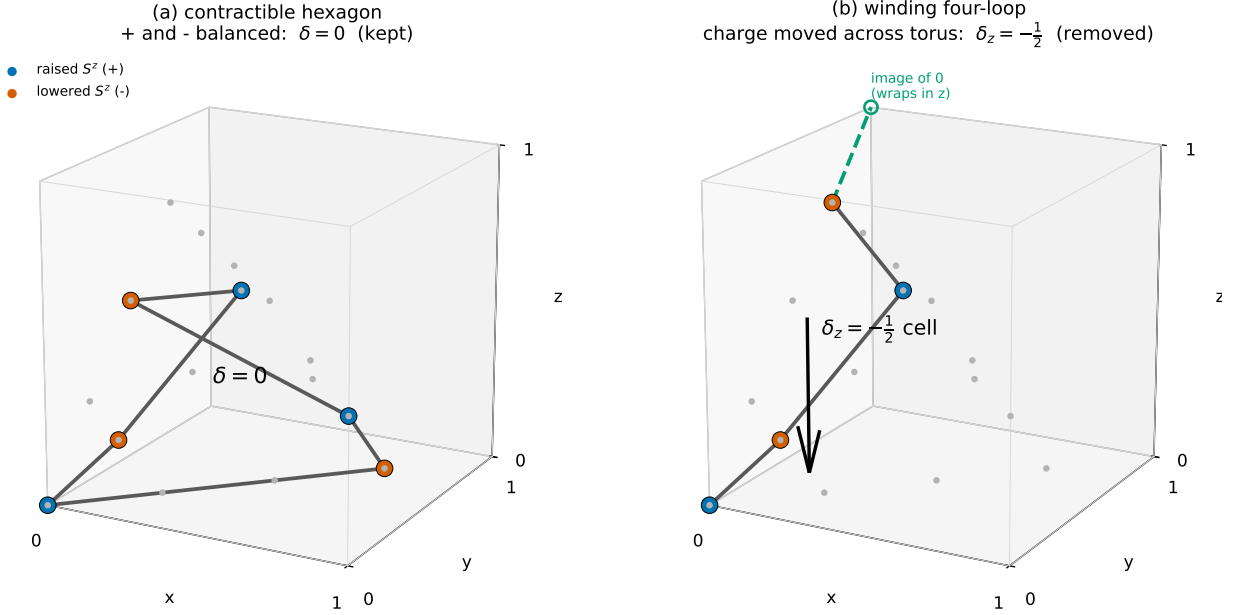


Figure 1: Charge transport on the 16-site cubic pyrochlore cell, $\delta = \sum_{\text{raised}} \mathbf{r} - \sum_{\text{lowered}} \mathbf{r}$. (a) A contractible hexagon (sites 0, 2, 7, 4, 14, 15) is balanced – the raised (+) and lowered (-) sites share a centroid – so $\delta = \mathbf{0}$ and it is kept. (b) A torus-winding four-loop (0, 2, 7, 5) closes only through the periodic boundary: the dashed bond runs to the periodic image of site 0, wrapping in z . Its raised and lowered charge are offset by half a cell, $\delta_z = -\frac{1}{2}$, so it is removed. Because the transported charge is half a lattice vector, the flux phase is $e^{-i\varphi/2}$ and must be swept *twice* (period 4π) to dephase it – the naive 2π leaves a residual.

- (ii) *Extended period.* A winding ring exchange transports emergent charge by *half* a lattice vector ($\mathbf{L}^{-1}\boldsymbol{\rho} = \frac{1}{2}$; verified for every winding four-loop and wrapping hexagon on both production clusters), so its transport phase $e^{-i\varphi \cdot (\mathbf{L}^{-1}\boldsymbol{\delta})} = e^{-i\varphi/2}$ winds at half the rate of φ and is φ -periodic only modulo 4π . Averaging over the naive $[0, 2\pi)^3$ therefore does *not* project (the phase makes half a turn, leaving a $\sim 40\%$ residual); the doubled period $[0, 4\pi)^3$ is required. (Equivalently, the flux quantum for the half-integer emergent charge is twice the 2π boundary-twist quantum.)

The average is at the *operator* level: one averages H_{eff} , then diagonalizes. Averaging an *observable* (a specific $C(T)$ or $S(\mathbf{q}, \omega)$) instead fails in any gauge (section 4).

4 Why it works

4.1 Ring exchanges are sums over virtual paths

The naive expectation is that a winding ring exchange picks up its holonomy $e^{-i\mathbf{w} \cdot \boldsymbol{\varphi}}$ and averages to zero. This is false in an instructive way.

Theorem 1 (Matched-pair factorization). *Let $\mathcal{C}$ be an ice-preserving cycle of length $2m$ and winding $\mathbf{w}$. At the leading order at which $O_{\mathcal{C}}$ appears, its coefficient in the boundary-gauge twisted $H_{\text{eff}}(\boldsymbol{\varphi})$*

is

$$c_C(\boldsymbol{\varphi}) = -\frac{g_C}{M} \sum_{\mu=1}^M e^{-i\mathbf{N}_\mu \cdot \boldsymbol{\varphi}}, \quad \sum_{\mu} \mathbf{N}_\mu = \mathbf{w} \quad (M=2), \quad (6)$$

summed over the $M \geq 2$ perfect matchings of $\mathcal{C}$ into disjoint NN dimer moves, $\mathbf{N}_\mu$ the wrap of matching μ . The gauge-invariant magnitude is $|c_C(\boldsymbol{\varphi})| = g_C |\cos(\frac{1}{2}\mathbf{w} \cdot \boldsymbol{\varphi})|$ for a two-matching loop.

Proof. At leading order O_C is generated only by flipping each of its $2m$ spins once, i.e. by a product of m disjoint NN dimer moves – a perfect matching; an even cycle has $M \geq 2$. Each hop $S_i^+ S_j^-$ carries $e^{-i\mathbf{n}_{ij} \cdot \boldsymbol{\varphi}}$, so matching μ carries $e^{-i\mathbf{N}_\mu \cdot \boldsymbol{\varphi}}$. For a 4-cycle the two matchings traverse each bond once, $\mathbf{N}_1 + \mathbf{N}_2 = \mathbf{w}$, giving $|1 + e^{-i\mathbf{w} \cdot \boldsymbol{\varphi}}| = 2|\cos(\frac{1}{2}\mathbf{w} \cdot \boldsymbol{\varphi})|$. $\square$

An operator thus lives in several transport sectors at once; the holonomy is only the *relative* phase of its matchings, and the magnitude *vanishes* at $\mathbf{w} \cdot \boldsymbol{\varphi} = \pi$ – which a single homology phase never does. In the boundary gauge a winding loop’s zero-wrap matching ($\mathbf{N}_\mu = \mathbf{0}$) is phase-free and survives any average, leaving $\frac{1}{2}g_C$. The smooth gauge removes this loophole because it weights a process by its transport $\boldsymbol{\delta}$ [eq. (4)], and *both* matchings of a winding loop transport charge.

4.2 The smooth-gauge average removes winding, keeps contractible

Theorem 2 (Transport selection). *Let $\ell_0 = \min_{\mathbf{w} \neq \mathbf{0}} |\mathbf{w} \cdot \mathbf{L}|$. If every ring exchange has dipole $|\boldsymbol{\rho}_C| < \ell_0$, then the extended-period smooth-gauge average (5) deletes every loop with $\boldsymbol{\rho}_C \neq \mathbf{0}$ at all orders in $J_\pm$ and keeps every loop with $\boldsymbol{\rho}_C = \mathbf{0}$ at full strength; on the production clusters these are exactly the winding and the contractible loops.*

Proof. The average keeps O_C iff $\boldsymbol{\delta}_C \in \mathbf{L}\mathbb{Z}^3$; since $\mathbf{N} \cdot \mathbf{L} \in \mathbf{L}\mathbb{Z}^3$, this is $\boldsymbol{\rho}_C \in \mathbf{L}\mathbb{Z}^3$, independent of order. With $|\boldsymbol{\rho}_C| < \ell_0$ the only lattice vector it can equal is $\mathbf{0}$. A genuine bulk plaquette (regular [111]-hexagon or its four-tetrahedron analogue) is centrosymmetric, so $\boldsymbol{\rho} = \mathbf{0}$ and it is kept. A winding loop has $\boldsymbol{\rho} \neq \mathbf{0}$: otherwise $\boldsymbol{\delta}_1 + \boldsymbol{\delta}_2 = \mathbf{w} \cdot \mathbf{L}$ with $|\mathbf{w} \cdot \mathbf{L}| \geq \ell_0$ would force some $|\boldsymbol{\delta}_\mu| \geq \frac{1}{2}\ell_0$ while $\boldsymbol{\delta}_\mu = \boldsymbol{\rho} + \mathbf{N}_\mu \mathbf{L}$ with $\boldsymbol{\rho} = \mathbf{0}$ makes $\boldsymbol{\delta}_\mu$ a lattice vector, hence $\mathbf{0}$ or $\geq \ell_0$ – contradiction. So winding loops are deleted. $\square$

On the production clusters the hypothesis holds with room to spare: the dipole of *every* four-loop and wrapping hexagon has magnitude $|\boldsymbol{\rho}| = a/\sqrt{2}$, below $\ell_0 = a$ (cubic) and $\sqrt{2}a$ (FCC), while every contractible hexagon has $\boldsymbol{\rho} = \mathbf{0}$ identically – the 0/0/1 pattern of table 1. The one honest bias: a *contractible* process with a nonzero dipole (possible only for non-centrosymmetric loops at order $J_\pm^4$ and beyond) would also be removed; within the NN model these are $O(J_\pm^4/J_{zz}^3)$, below g_{hex} .

4.3 Why the naive versions fail

Corollary 1. (a) Boundary gauge: *the operator average keeps $\frac{1}{2}g_C$ (the zero-wrap matching is phase-free).* (b) Naive period: *the smooth-gauge average over $[0, 2\pi)^3$ leaves a residual, because $\mathbf{L}^{-1}\boldsymbol{\rho} = \frac{1}{2}$ is half-integer; the doubled period $[0, 4\pi)^3$ is required.* (c) Observable average: *averaging $\mathcal{O}(g)$ rather than H_{eff} keeps every even power of the gauge-invariant $|c_C| = g_C |\cos(\frac{1}{2}\mathbf{w} \cdot \boldsymbol{\varphi})|$, so the spurious scale survives in any gauge.*

5 Computational verification

Each claim is checked exactly on the 16-site cluster.

(1) Matched-pair magnitude. For a winding four-loop [sites $(0, 2, 8, 10)$, $\mathbf{w} = (-1, 0, -1)$] the exact second-order coefficient obeys, along $\mathbf{w}$, $|c(\boldsymbol{\varphi})|/g_4 = 1.000, 0.866, 0.707, 0.500, 0.000, \dots = |\cos(\frac{1}{2}\mathbf{w} \cdot \boldsymbol{\varphi})|$ to all digits, vanishing at $\mathbf{w} \cdot \boldsymbol{\varphi} = \pi$.

(2) The average, gauge and period. The winding four-loop coefficient under the operator-level average is

average	coeff. $/g_4$	$T_{\text{peak}}/g_{\text{hex}}$
none (bare)	-1.00	3.18
boundary gauge	-0.50	1.63
smooth gauge, $[0, 2\pi)^3$	+0.40	0.84
smooth gauge, $[0, 4\pi)^3$ (extended)	0.00	0.62

while a contractible hexagon is $+g_{\text{hex}}$ in every case (survival 0/0/1, table 1, to machine precision). Only the uniform gauge over the extended period removes the loop; both naive choices leave a residual.

(3) Two-peak specific heat. Figure 2 shows the full $C(T)$ from exact 16-site diagonalization (12870 states in $S_{\text{tot}}^z = 0$), with the gauge band cleaned by the translationally-invariant average and the spinon sector taken from the same spectrum. The low- T gauge (photon) peak moves from the artifact scale g_4 (bare) to the photon scale g_{hex} under the average, while the high- T charge (spinon) peak at $T \sim J_{zz}/3$ is essentially unchanged – as it must be, being uncontaminated physics outside the ice manifold. Averaging the observable instead barely moves the gauge peak (corollary 1(c)).

6 Demonstration and outlook

The gauge peak of the exact 32-site FCC H_{eff} (full spectrum of the 2970-state ice manifold), in units of g_{hex} : bare periodic ED sits at the artifact scale and drifts up as $g_4/g_{\text{hex}} = J_{zz}/3|J_{\pm}|$ toward weak coupling (5.4, 3.1 at $J_{\pm} = -0.03, -0.05$), while the translationally-invariant average brings it to $O(1)$ (0.6, 0.3). The same holds for the emergent-photon structure factor $S^{zz}(\mathbf{q}, \omega) = \sum_n |\langle n | S_{\mathbf{q}}^z | 0 \rangle|^2 \delta(\omega - \omega_n)$, whose low- ω weight moves from $\sim g_4$ (bare) to $\sim g_{\text{hex}}$ (averaged) on both clusters and both flux signs.

The couplings scale as $J_{\pm}^k$ exactly, so one effective-Hamiltonian build serves every coupling and both signs. The program is: (A) benchmark the averaged 0-flux $C(T)$, $S(T)$ and photon S^{zz} against sign-problem-free QMC at a common coupling; (B) cross into π -flux and compute the same quantities where QMC cannot; (C) evaluate at the measured exchange parameters of $\text{Ce}_2\text{Zr}_2\text{O}_7$, $\text{Ce}_2\text{Sn}_2\text{O}_7$, $\text{Ce}_2\text{Hf}_2\text{O}_7$ and confront the neutron $S(\mathbf{q}, \omega)$ and the specific heat. The uncontaminated spinon continuum is supplied by full-Hilbert-space finite-temperature Lanczos, with twist averaging there used for its standard purpose – Brillouin-zone interpolation of the genuine bands.

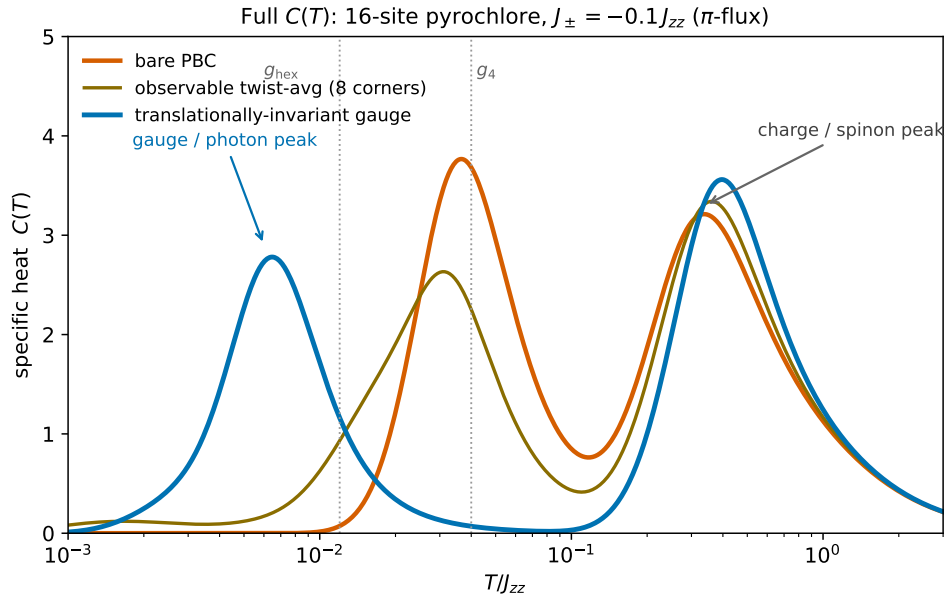


Figure 2: Full specific heat of the 16-site π -flux cluster ($J_{\pm} = -0.1 J_{zz}$). Bare periodic ED (orange) places the low- T gauge peak at the finite-size artifact scale g_4 ; averaging the *observable* over eight corners (olive) barely moves it; translationally-invariant twist averaging of the *Hamiltonian* (blue) moves it to the photon scale g_{hex} . The high- T charge (spinon) peak at $T \sim J_{zz}/3$ is genuine, uncontaminated, and unchanged.